



Deep Learning

Dr. Mehran Safayani

safayani@iut.ac.ir

safayani.iut.ac.ir



<https://www.aparat.com/mehran.safayani>

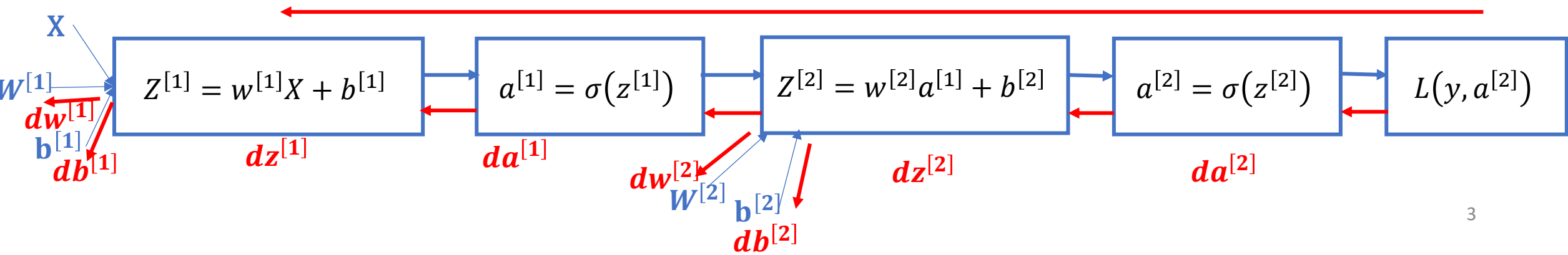
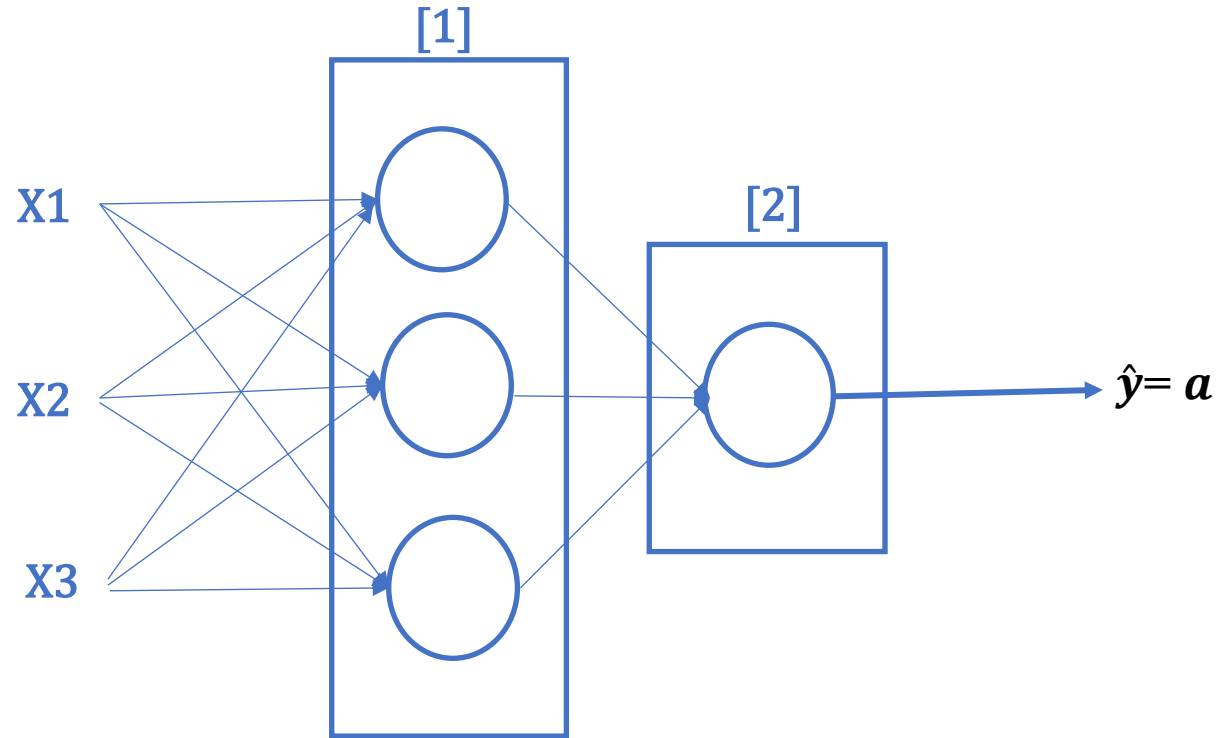


https://github.com/safayani/deep_learning_course

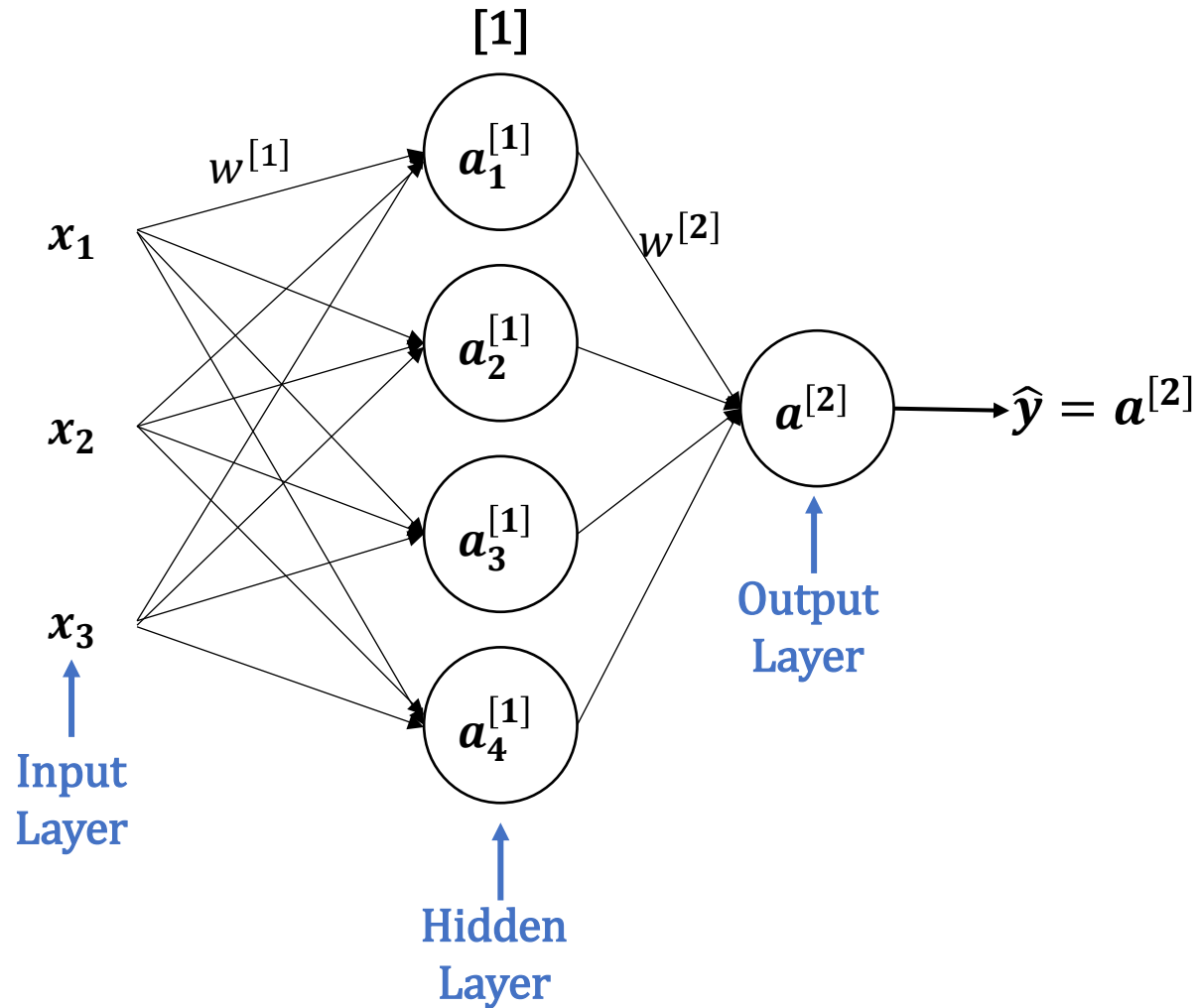


Neural Network Representation

Neural Networks



Neural Network Representation



2 layer NN

$$w^{[1]}, b^{[1]}$$

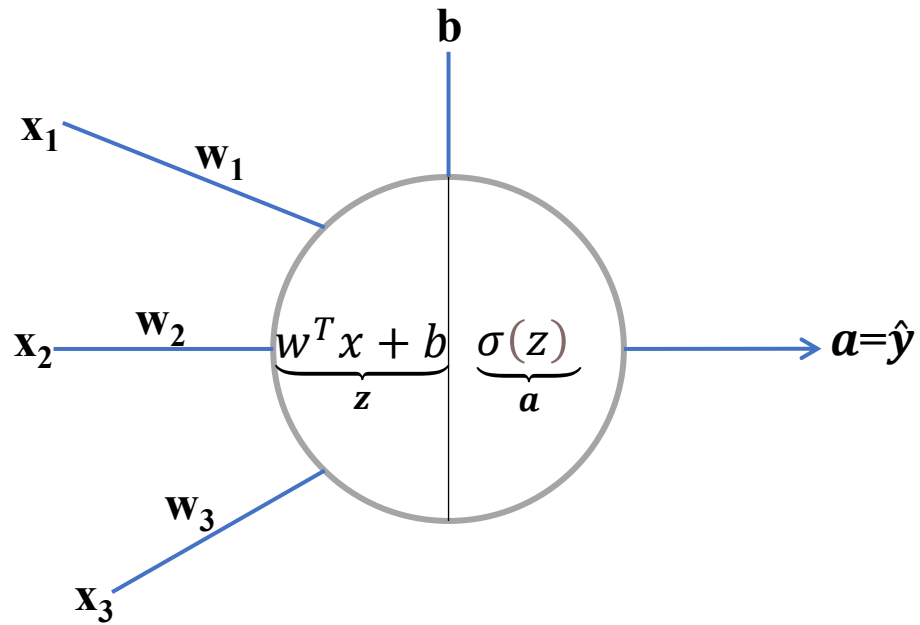
$$(4 \times 3) (4 \times 1)$$

$$w^{[2]}, b^{[2]}$$

$$(1,4) (1,1)$$

$$a = \begin{bmatrix} a_1^{[1]} \\ a_2^{[1]} \\ a_3^{[1]} \\ a_4^{[1]} \end{bmatrix}$$

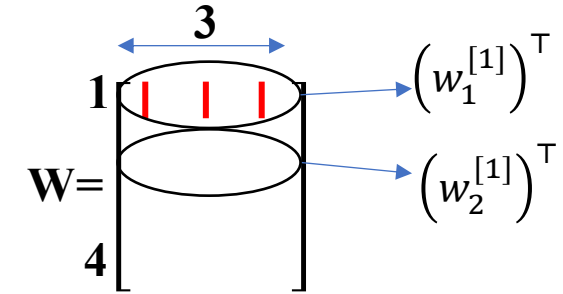
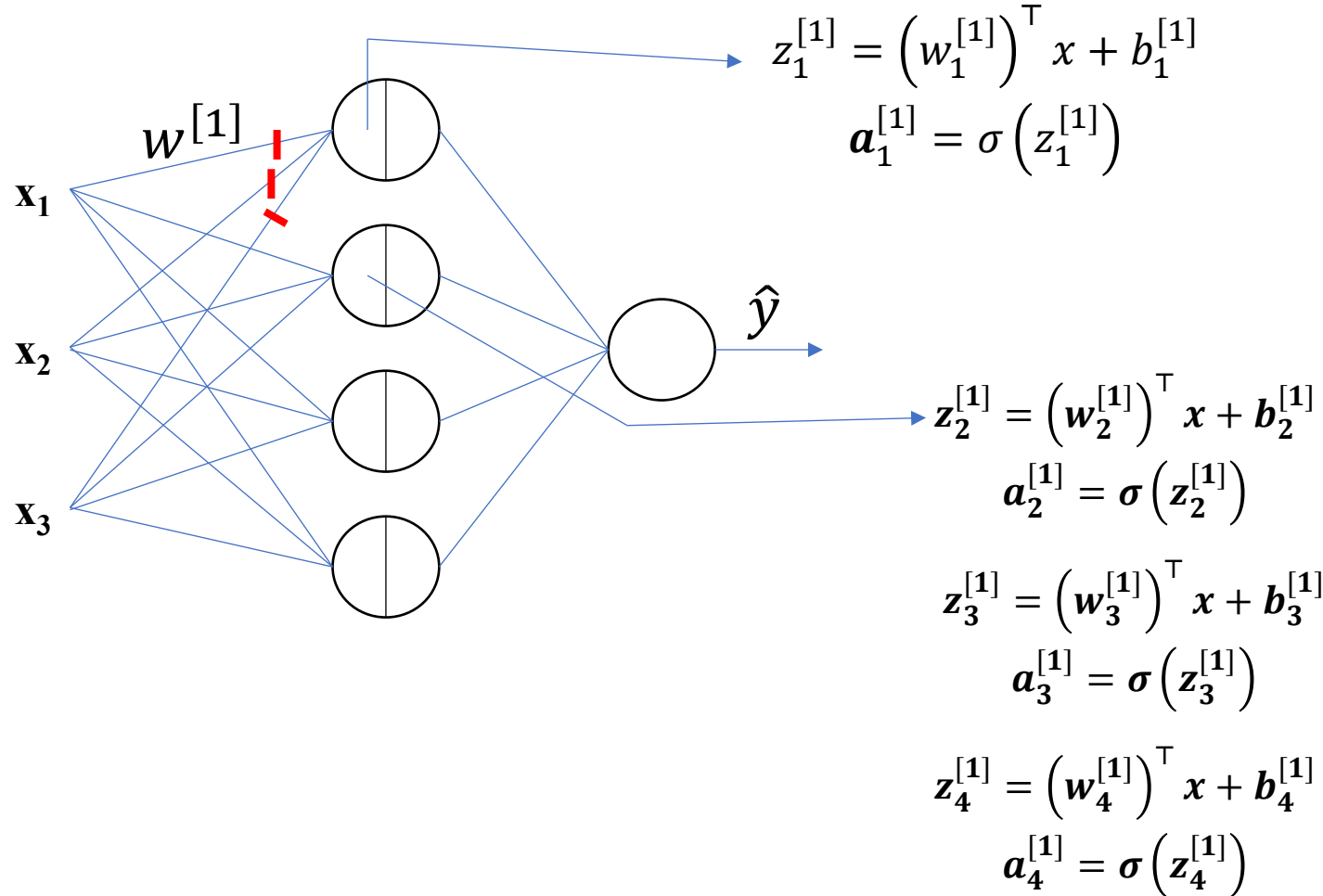
Neural Network Representation



$$z = w^T x + b$$

$$a = \sigma(z)$$

Neural Network Representation



Neural Network Representation

$$\bullet Z^{[1]} = \underset{\substack{(4,3)}}{\begin{bmatrix} w_1^{[1]T} \\ w_2^{[1]T} \\ w_3^{[1]T} \\ w_4^{[1]T} \end{bmatrix}} \underset{\substack{(3,1)}}{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}} + \underset{\substack{(4,1)}}{\begin{bmatrix} b_1^{[1]} \\ b_2^{[1]} \\ b_3^{[1]} \\ b_4^{[1]} \end{bmatrix}} = \begin{bmatrix} w_1^{[1]T}x + b_1^{[1]} \\ w_2^{[1]T}x + b_2^{[1]} \\ w_3^{[1]T}x + b_3^{[1]} \\ w_4^{[1]T}x + b_4^{[1]} \end{bmatrix} = \begin{bmatrix} z_1^{[1]} \\ z_2^{[1]} \\ z_3^{[1]} \\ z_4^{[1]} \end{bmatrix}$$

$$\bullet a^{[1]} = \begin{bmatrix} a_1^{[1]} \\ a_2^{[1]} \\ a_3^{[1]} \\ a_4^{[1]} \end{bmatrix} = \sigma(z^{[1]}) = \begin{bmatrix} \sigma(z_1^{[1]}) \\ \sigma(z_2^{[1]}) \\ \sigma(z_3^{[1]}) \\ \sigma(z_4^{[1]}) \end{bmatrix}$$

Neural Network Representation

- $z^{[1]}_{4 \times 1} = w^{[1]}_{4 \times 3} \tilde{x}^{a^{[0]}}_{3 \times 1} + b^{[1]}_{4 \times 1}$
- $a^{[1]}_{4 \times 1} = \sigma(z^{[1]}_{4 \times 1})$
- $z^{[2]}_{1 \times 1} = w^{[2]}_{1 \times 4} a^{[1]}_{4 \times 1} + b^{[2]}_{1 \times 1}$
- $a^{[2]}_{1 \times 1} = \sigma(z^{[2]}_{1 \times 1})$
- For $i=1$ to m

$$z^{[1]}(i) = w^{[1]}x^{(i)} + b^{[1]}$$

$$a^{[1]}(i) = \sigma(z^{[1]}(i))$$

$$z^{[2]}(i) = w^{[2]}a^{[1]}(i) + b^{[2]}$$

$$a^{[2]}(i) = \sigma(z^{[2]}(i))$$

Neural Network Representation

- $X = \begin{bmatrix} x^{(1)} & x^{(2)} & \dots & \dots & x^{(m)} \end{bmatrix}_{(nx, m)}$

- $Z^{[1]}_{4 \times m} = w^{[1]}_{4 \times 3} X_{3 \times m} + \underbrace{b^{[1]}_{4 \times 1}}_{\text{broadcasting}}$

- $A^{[1]}_{4 \times m} = \sigma(Z^{[1]}_{4 \times m})$

- $Z^{[2]}_{1 \times m} = w^{[2]}_{1 \times 4} A^{[1]}_{4 \times m} + b^{[2]}_{1 \times 1}$

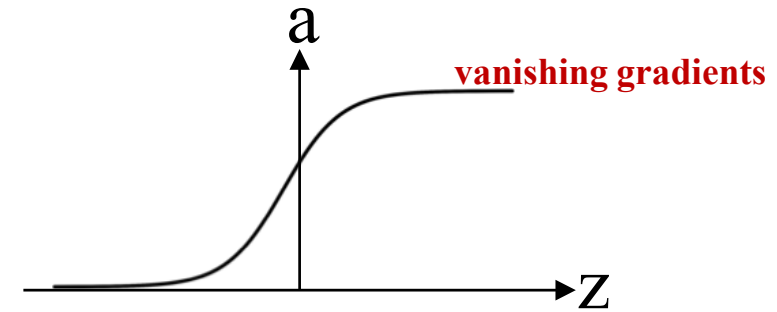
- $A^{[2]}_{1 \times m} = \sigma(Z^{[2]}_{1 \times m})$

$$Z^{[1]} = \begin{bmatrix} z^{1} & z^{[1](2)} & \dots & \dots & z^{[1](m)} \end{bmatrix}$$

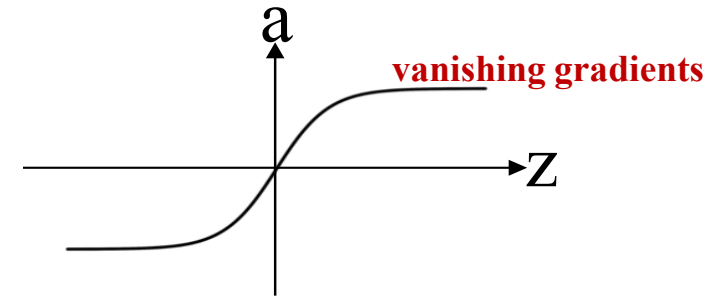
$$A^{[1]} = \begin{bmatrix} a^{1} & a^{[1](2)} & \dots & \dots & a^{[1](m)} \end{bmatrix}$$

Activation function

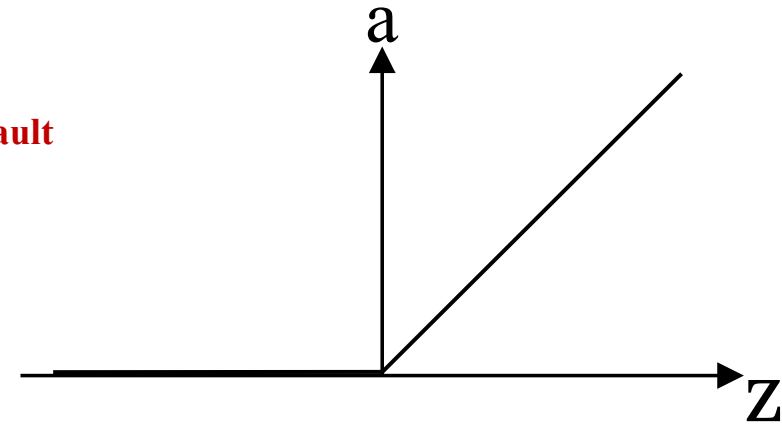
- $\text{sigmoid}(z) = \frac{1}{1+e^{-z}}$ **don't be used except for output**



- $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$ **don't be used except for output**

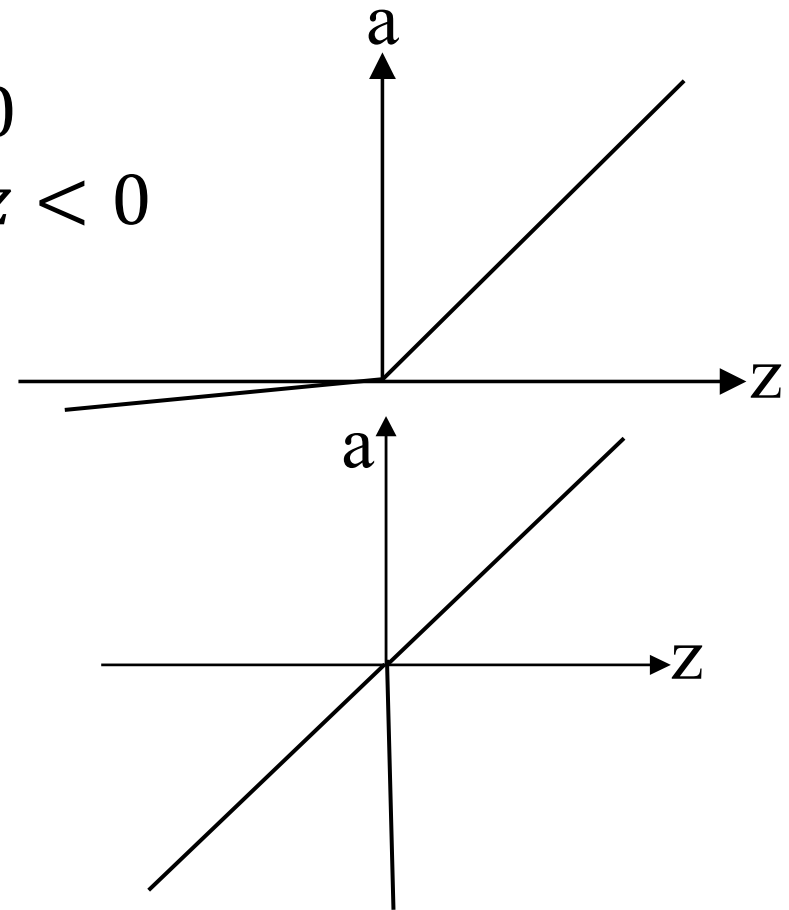


- $\text{Relu}(z) = \max(0, z) = \begin{cases} z & z \geq 0 \\ 0 & z < 0 \end{cases}$ **default**



Activation function

- $Leaky\ Relu(z) = \max(0, z) = \begin{cases} z & z \geq 0 \\ 0.01z & z < 0 \end{cases}$



- $Linear(z) = z$

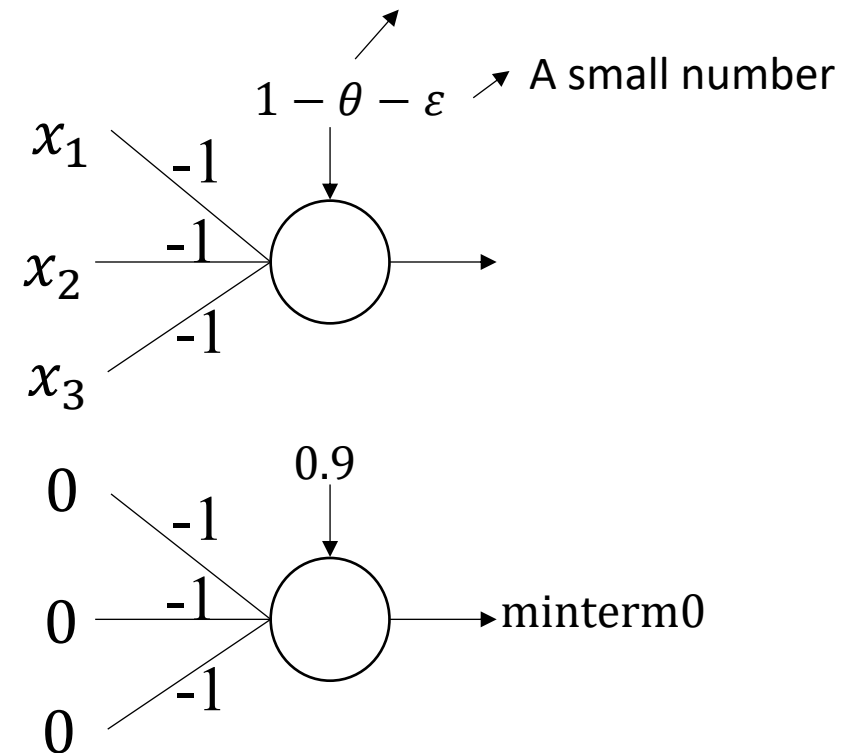
Threshold Logic Unit(TLU)

- $z = w^T x + b$

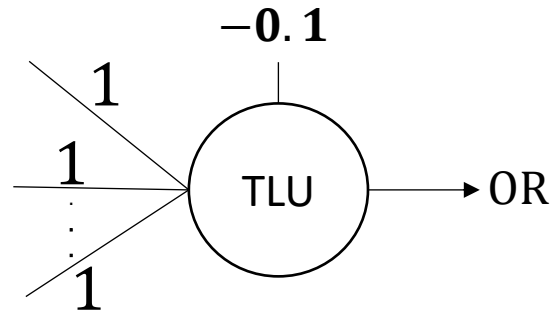
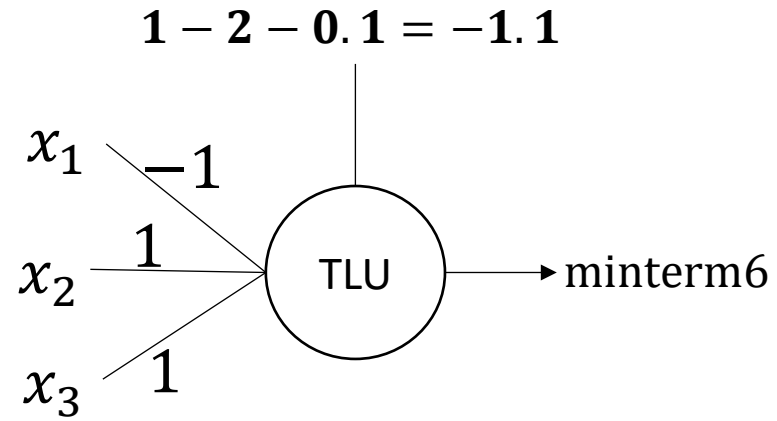
- $\hat{y} = \begin{cases} 1 & w^T x + b \geq 0 \\ 0 & w^T x + b < 0 \end{cases}$

- | x_1 | x_2 | x_3 | y | |
|-------|-------|-------|-----|---------------------------------|
| 0 | 0 | 0 | 1 | $\bar{x}_1 \bar{x}_2 \bar{x}_3$ |
| 0 | 0 | 1 | 0 | |
| 0 | 1 | 0 | 0 | |
| . | . | . | . | |
| . | . | . | . | |
| 0 | 1 | 1 | 1 | $\bar{x}_1 x_2 x_3$ |

Number of none zero items in the minterm

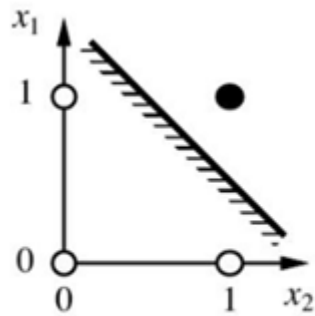


Threshold Logic Unit(TLU)

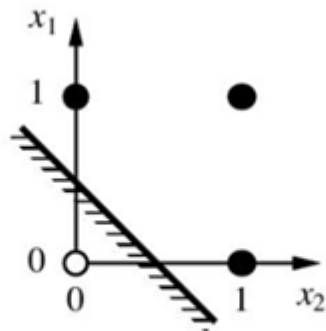


And, Or, XOR problem

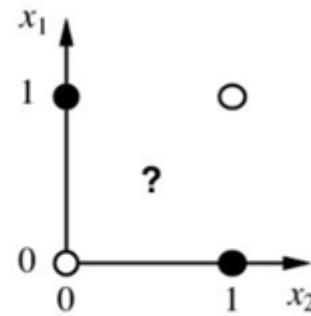
x1	x2	and	or	xor
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0



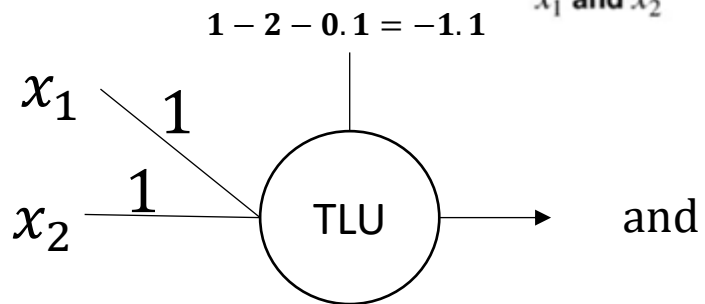
x_1 and x_2



x_1 or x_2

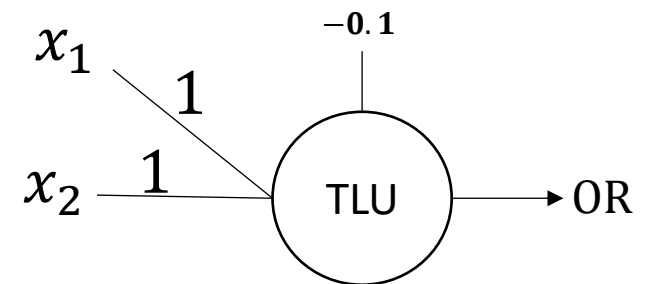


x_1 xor x_2



$$x_1 + x_2 - 1.1 = 0$$

$$x_1 = -x_2 + 1.1$$

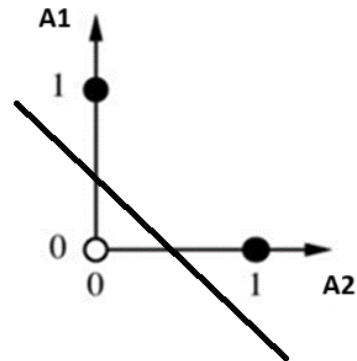
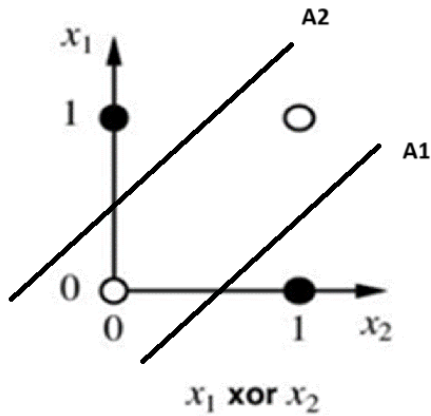


$$x_1 + x_2 - 0.1 = 0$$

$$x_1 = -x_2 + 0.1$$

And, Or, XOR problem

x_1	x_2	$A1 = x_1'x_2$	$A2 = x_1x_2'$	xor
0	0	0	0	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0



$$\begin{aligned} -x_1 + x_2 - 0.1 &= 0 & x_1 - x_2 - 0.1 &= 0 \\ x_1 &= x_2 - 0.1 & x_1 &= x_2 + 0.1 \end{aligned}$$

$$\begin{aligned} a_1 + a_2 - 0.1 &= 0 \\ a_1 &= -a_2 + 0.1 \end{aligned}$$

